

1] The components of the mass flow vector $\rho \vec{u}$ are $\rho u_x = 4x^2y$, $\rho u_y = xyz$, and $\rho u_z = yz^2$.

- (a) Compute the net mass outflow through the closed surface formed by the planes, $x = 0$, $x = 1$, $y = 0$, $y = 1$, $z = 0$, $z = 1$.
- (b) Compute $\vec{\nabla} \cdot (\rho \vec{u})$ and integrate over the volume bounded by the surface defined in part a).
- (c) Explain why the results in a) and b) should, or should not, be equal.

4.) $\rho \vec{u} = \langle 4x^2y, xyz, yz^2 \rangle$

$x=0$:

$$\begin{aligned} \Phi &= \int_{y=0}^{y=1} \int_{z=0}^{z=1} \langle 4x^2y, xyz, yz^2 \rangle \cdot \langle -1, 0, 0 \rangle^T dz dy \\ &= \int_{y=0}^{y=1} \int_{z=0}^{z=1} -4x^2y dz dy = \boxed{0} \end{aligned}$$

$x=1$:

$$\begin{aligned} \Phi &= \int_{y=0}^{y=1} \int_{z=0}^{z=1} \langle 4x^2y, xyz, yz^2 \rangle \cdot \langle 1, 0, 0 \rangle^T dz dy \\ &= \int_{y=0}^{y=1} \int_{z=0}^{z=1} 4x^2y dz dy = 4 \int_{y=0}^{y=1} xy dy = \boxed{2} \end{aligned}$$

$y=0$:

$$\begin{aligned} \Phi &= \int_{x=0}^{x=1} \int_{z=0}^{z=1} \langle 4x^2y, xyz, yz^2 \rangle \cdot \langle 0, -1, 0 \rangle^T dz dx \\ &= \int_{x=0}^{x=1} \int_{z=0}^{z=1} -xyz dz dx = \boxed{0} \end{aligned}$$

$$\underline{y=1:}$$

$$\begin{aligned}\Phi &= \int_{x=0}^{x=1} \int_{z=0}^{z=1} \langle 4x^2y, xyz, yz^2 \rangle \cdot \langle 0, 1, 0 \rangle^T dz dx \\ &= \int_{x=0}^{x=1} \int_{z=0}^{z=1} xyz = \frac{1}{2} \int_{x=0}^{x=1} x dx = \boxed{\frac{1}{4}}\end{aligned}$$

$$\underline{z=0:}$$

$$\begin{aligned}\Phi &= \int_{x=0}^{x=1} \int_{y=0}^{y=1} \langle 4x^2y, xyz, yz^2 \rangle \cdot \langle 0, 0, -1 \rangle^T dy dx \\ &= \int_{x=0}^{x=1} \int_{y=0}^{y=1} -yz^2 dy dz = \boxed{0}\end{aligned}$$

$$\underline{z=1:}$$

$$\begin{aligned}\Phi &= \int_{x=0}^{x=1} \int_{y=0}^{y=1} \langle 4x^2y, xyz, yz^2 \rangle \cdot \langle 0, 0, 1 \rangle^T dy dx \\ &= \int_{x=0}^{x=1} \int_{y=0}^{y=1} yz^2 dy dx = \frac{1}{2} \int_{x=0}^{x=1} dx = \boxed{\frac{1}{2}}\end{aligned}$$

Sum of the mass fluxes through each plane: $\boxed{2^{3/4}}$

B.)

$$\begin{aligned}\nabla \cdot (\rho \underline{u}) &= \partial_x(4x^2y) + \partial_y(xyz) + \partial_z(yz^2) \\ &= 8xy + xz + 2yz\end{aligned}$$

$$\int_{x=0}^{x=1} \int_{y=0}^{y=1} \int_{z=0}^{z=1} (8xy + xz + 2yz) dz dy dx$$

$$= \int_{x=0}^{x=1} \int_{y=0}^{y=1} \left(8xyz + \frac{1}{2}xz^2 + yz^2 \right) \Big|_{z=0}^{z=1} dy dx$$

$$= \int_{x=0}^{x=1} \int_{y=0}^{y=1} \left(8xy + \frac{1}{2}x + y \right) dy dx$$

$$= \int_{x=0}^{x=1} \left(4xy^2 + \frac{1}{2}xy + \frac{1}{2}y^2 \right) \Big|_{y=0}^{y=1} dx$$

$$= \int_{x=0}^{x=1} \left(4x + \frac{1}{2}x + \frac{1}{4} \right) dx = \left(2x^2 + \frac{1}{4}x^2 + \frac{1}{2}x \right) \Big|_{x=0}^{x=1} = \boxed{2 \frac{3}{4}}$$

c.) The results in a.) and b.) should be the same, given by the divergence theorem. ///

2] For the following flows find the missing velocity component needed for the flow to satisfy the incompressible continuity equation

(a) $u_1 = x_1^2 + x_2^2 + a^2$, $v_2 = -x_1x_2 - x_2x_3 - x_1x_3$, $u_3 = ?$ ↗ Should be u_2

(b) $u_1 = \ln(x_2^2 + x_3^2)$, $u_2 = \sin(x_1^2 + x_3^2)$, $u_3 = ?$

(c) $u_1 = ?$, $u_2 = \frac{x_2}{(x_1^2 + x_2^2 + x_3^2)^{3/2}}$, $u_3 = \frac{x_3}{(x_1^2 + x_2^2 + x_3^2)^{3/2}}$

A.) $\nabla \cdot \underline{u} = \frac{\partial}{\partial x_1}(x_1^2 + x_2^2 + a^2) + \frac{\partial}{\partial x_2}(-x_1x_2 - x_2x_3 - x_1x_3) + \frac{\partial}{\partial x_3}(u_3) = 0$

$$\Rightarrow 2x_1 - x_1 - x_3 + \frac{\partial u_3}{\partial x_3} = 0 \Rightarrow \frac{\partial u_3}{\partial x_3} = x_1 + x_3 - 2x_1$$

$$\Rightarrow \int du_3 = \int (x_3 - x_1) dx_3 \Rightarrow u_3 = \frac{1}{2}x_3^2 - x_1x_3 + f(x_1, x_2)$$

B.) $\nabla \cdot \underline{u} = \frac{\partial}{\partial x_1}[\ln(x_2^2 + x_3^2)] + \frac{\partial}{\partial x_2}[\sin(x_1^2 + x_3^2)] + \frac{\partial}{\partial x_3}(u_3) = 0$

$$\Rightarrow 0 + 0 + \frac{\partial u_3}{\partial x_3} = 0 \Rightarrow u_3 = f(x_1, x_2)$$

C.) $\nabla \cdot \underline{u} = \frac{\partial}{\partial x_1}(u_1) + \frac{\partial}{\partial x_2}\left[\frac{x_2}{(x_1^2 + x_2^2 + x_3^2)^{3/2}}\right] + \frac{\partial}{\partial x_3}\left[\frac{x_3}{(x_1^2 + x_2^2 + x_3^2)^{3/2}}\right] = 0$

$$\text{Let } r = (x_1^2 + x_2^2 + x_3^2)^{1/2} \Rightarrow r^2 = x_1^2 + x_2^2 + x_3^2$$

$$\Rightarrow \frac{\partial u_1}{\partial x_1} + \frac{r^2 - 3x_2^2}{r^5} + \frac{r^2 - 3x_3^2}{r^5} = 0$$

$$\Rightarrow \frac{\partial u_1}{\partial x_1} = \frac{3(r^2 - x_1^2) - 2r^2}{r^5} = \frac{r^2 - 3x_1^2}{r^5}$$

$$\Rightarrow \int du_1 = \int \frac{r^2 - 3x_1^2}{r^5} dx_1 \Rightarrow u_1 = \frac{x_1}{(x_1^2 + x_2^2 + x_3^2)^{3/2}} + f(x_2, x_3)$$

3] Consider the velocity field $\mathbf{u} = (\alpha x/t, 0, 0)$ (in Cartesian), where α is a constant. The density of the fluid is spatially uniform, but time-dependent, i.e., ρ is a function of t .

- (a) Determine the density field $\rho(t)$, given $\rho = \rho_0$ at $t = t_0$.
- (b) What are the temporal $(\partial \mathbf{u} / \partial t)$, advective $(\mathbf{u} \cdot (\nabla \mathbf{u}))$, and total $(D\mathbf{u}/Dt)$ accelerations for this flow field? What does $\alpha = 1$ imply?

A.) $\underline{u} = \begin{bmatrix} \frac{\alpha x}{t} \\ 0 \\ 0 \end{bmatrix}, \alpha \in \mathbb{R}, \rho = \rho(t)$

$$\partial_t \rho + \nabla \cdot (\rho \underline{u}) = 0 \Rightarrow \partial_t \rho + [\partial_x (\rho u_x) + \partial_y (\rho u_y) + \partial_z (\rho u_z)] = 0$$

$$\Rightarrow \partial_t \rho + \rho \left[\partial_x \left(\alpha \frac{x}{t} \right) + 0 + 0 \right] = 0 \Rightarrow \partial_t \rho = -\rho \frac{\alpha}{t}$$

$$\Rightarrow \int_{t_0}^t \rho^{-1} d\rho = -\alpha \int_{t_0}^t t^{-1} dt \Rightarrow \ln(\rho) \Big|_{t_0}^t = -\alpha \ln(t) \Big|_{t_0}^t$$

$$\Rightarrow \ln(\rho(t)) - \ln(\rho(t_0)) = -\alpha (\ln(t) - \ln(t_0))$$

$$\Rightarrow \ln\left(\frac{\rho}{\rho_0}\right) = -\alpha \ln\left(\frac{t}{t_0}\right) \Rightarrow \boxed{\rho(t) = \rho_0 \left(\frac{t_0}{t}\right)^\alpha}$$

B.)